1. Write an assembly language program that takes 2 single-digit decimal inputs (x & y) from the user, calculates the result as (x + y - 3), and prints the result.

Sample Input & Output:

```
Enter 1st decimal number, x: 5
Enter 2nd decimal number, y: 6
Result (x+2+y-3) in decimal: 10
```

2. Write a program to read three lowercase characters from the user and then display them in UPPERCASE in the same (forward) order, each on a separate line.

Sample Input & Output:

```
Enter three characters: abc
A
B
C
```

3. Write an assembly language program that reads three single-digit decimal numbers from the user and displays their sum. The result may be a two-digit number — handle the output accordingly.

Sample Input & Output:

```
Enter 1st digit: 7
Enter 2nd digit: 8
Enter 3rd digit: 9
Sum of digits: 24
```

4. Write a program to read two uppercase characters from the user and display each character together with its lowercase equivalent on the same line, for each character on a separate line.

Sample Input & Output:

```
Enter two uppercase characters: MK
Mm
Kk
```

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Answer to the question no 1.

```
.MODEL SMALL

.STACK 100H

.DATA

MSG1 DB "Enter 1st decimal number, x: $"

MSG2 DB 0DH,0AH,"Enter 2nd decimal number, y: $"

MSG3 DB 0DH,0AH,"Result (x+2+y-3) in decimal: $"

X DB ?

Y DB ?

RESULT DB ?
```

```
.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX
    LEA DX, MSG1
    MOV AH, 09H
    INT 21H
    MOV AH, 01H
    INT 21H
    SUB AL, 30H
    MOV X, AL
    LEA DX, MSG2
    MOV AH, 09H
    INT 21H

    MOV AH, 01H
    INT 21H
    SUB AL, 30H
    MOV Y, AL

    MOV AL, X
    ADD AL, 2
    ADD AL, Y
    SUB AL, 3

    MOV RESULT, AL
```

```
    LEA DX, MSG3
    MOV AH, 09H
    INT 21H

    MOV AL, RESULT
    ADD AL, 30H
    MOV DL, AL
    MOV AH, 02H
    INT 21H
    MOV AH, 4CH
    INT 21H

MAIN ENDP
END MAIN
```

Answer to the question no 2

```
.MODEL SMALL
.STACK 100H
.DATA
MSG DB "Enter three characters: $"
.CODE
MAIN PROC
    MOV AX, @DATA
    MOV DS, AX

    LEA DX, MSG
```

```
MOV AH, 09H
INT 21H

MOV AH, 01H
INT 21H
SUB AL, 20H
MOV DL, AL

MOV AH, 02H
INT 21H
MOV DL, 0DH
MOV AH, 02H
INT 21H
MOV DL, 0AH
INT 21H

MOV AH, 01H
INT 21H
SUB AL, 20H
MOV DL, AL

MOV AH, 02H
INT 21H

MOV DL, 0DH
MOV AH, 02H
```

```
    INT 21H
    MOV DL, 0AH
    INT 21H
    MOV AH, 01H
    INT 21H
    SUB AL, 20H
    MOV DL, AL

    MOV AH, 02H
    INT 21H

    MOV AH, 4CH
    INT 21H

MAIN ENDP
END MAIN
```

Answer to the question no 3

```
.MODEL SMALL
.STACK 100H
.DATA
MSG1 DB "Enter 1st digit: $"
MSG2 DB 0DH,0AH,"Enter 2nd digit: $"
MSG3 DB 0DH,0AH,"Enter 3rd digit: $"
MSG4 DB 0DH,0AH,"Sum of digits: $"
D1 DB ?
```

```
D2 DB ?
D3 DB ?
SUM DB ?

.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX

    LEA DX,MSG1
    MOV AH,09H
    INT 21H

    MOV AH,01H
    INT 21H
    SUB AL,30H
    MOV D1,AL

    LEA DX,MSG2
    MOV AH,09H
    INT 21H

    MOV AH,01H
    INT 21H
    SUB AL,30H
    MOV D2,AL
```

```
LEA DX,MSG3
MOV AH,09H
INT 21H

MOV AH,01H
INT 21H
SUB AL,30H
MOV D3,AL

MOV AL,D1
ADD AL,D2
ADD AL,D3
MOV SUM,AL

LEA DX,MSG4
MOV AH,09H
INT 21H
MOV AL,SUM
MOV AH,0
MOV BL,10
DIV BL

MOV CL,AH

MOV DL,AL
```

```
    ADD DL,30H
    MOV AH,02H
    INT 21H

    MOV DL,CL
    ADD DL,30H
    MOV AH,02H
    INT 21H
    MOV AH,4CH
    INT 21H

MAIN ENDP
END MAIN
```

Answer to the question no 4

```
.MODEL SMALL
.STACK 100H

.DATA
MSG DB "Enter two uppercase characters: $"
.CODE
MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    LEA DX,MSG
```

```
MOV AH,09H
INT 21H
MOV AH,01H
INT 21H
MOV BL,AL
MOV DL,BL
MOV AH,02H
INT 21H

ADD BL,20H
MOV DL,BL
MOV AH,02H
INT 21H

MOV DL,0DH
MOV AH,02H
INT 21H
MOV DL,0AH
INT 21H

MOV AH,01H
INT 21H
MOV BL,AL

MOV DL,BL
MOV AH,02H
```

```
    INT 21H

    ADD BL,20H
    MOV DL,BL
    MOV AH,02H
    INT 21H

    MOV AH,4CH
    INT 21H

MAIN ENDP
END MAIN
```

Output Trace Practice Questions

Answer to the question no 1

Register	Final Value (Hex)	Final Value (Decimal)
AX	**000Dh**	**13**
BX	**0002h**	**2**
CX	**000Bh**	**11**

Answer to the question no2

Register	Final Value (Hex)
AL	FFh
BL	06h

Answer to the question no3

A

EQU Name	Hex Value	Decimal Value
BASE_VAL	0Ah	10
STEP	03h	3
LIMIT	0Dh	13

B

Step	Instruction	AL	BL	BH	CL
(1)	MOV AL, BASE_VAL	0Ah	—	—	—
(2)	MOV BL, STEP	0Ah	03h	—	—
(3)	ADD AL, BL	0Dh	03h	—	—
(4)	MOV CL, LIMIT	0Dh	03h	—	0Dh
(5)	SUB AL, CL	00h	03h	—	0Dh
(6)	ADD CL, STEP	00h	03h	—	10h
(7)	MOV BH, AL	00h	03h	00h	10h

Answer to the question no4

Line #	Output Text Printed
1	Hi
2	Hi
3	Hi
4	Bye

Answer to the question no5

Scenario	AL value	CL value	Output Printed
A	03h	05h	Less
B	07h	05h	Greater
C	05h	05h	Equal
D	FFh (−1)	05h	Less

Answer to the question no6

Register	Hex	Decimal
AX	0Ah	10
CX	00h	0